

Paolo Olivucci

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github.com/polivucci | [Google Scholar](#)

Researcher with a PhD in computational fluid mechanics and 4+ years of postdoctoral experience building physics-oriented ML models and optimization frameworks for aerodynamics and fluid flows. Background comprises high-performance computing, CFD, turbulent flow control, and translating computational and statistics research into predictive tools that improve efficiency and offer physical insight. Looking to bring this to computational modelling and data science roles in the aerospace, applied sciences, or financial sectors.

TECHNICAL SKILLS

Programming Languages: Python, Fortran, MATLAB

Machine Learning & Statistics: PyTorch, JAX, NumPy, Pandas, scikit-learn, GPyTorch, BoTorch

CFD & Simulation: PyFR (FEM), Incompact3D (FD), OpenFOAM (FV), XFOil

HPC & Infrastructure: Linux, Git, Singularity, SLURM

EXPERIENCE

Postdoctoral Research Associate

2022 – Present

TU Braunschweig

Germany

Multi-fidelity probabilistic ML models to predict costly aerodynamic quantities with calibrated uncertainty.

Cluster-based generative models for reduced-order dynamical predictions at out-of-data regimes.

Machine learning summarization of load measurements induced by atmospheric turbulence on delta-wing platforms.

Lecturer – “Smart Sensing in Fluid Dynamics”

2025 – 2026

TU Braunschweig

Germany

Designed and co-delivered a graduate-level course. Topics: experiment design, active learning, uncertainty modelling, sparse sensing, information fusion. [\[syllabus\]](#)

Doctoral Researcher

2016 – 2021

University of Sheffield

UK

Simulated active and passive flow control through rotating discs, demonstrating passive frictional drag reduction of 5%.

Simulated fully resolved fluid-body dynamics for solid discs in turbulent channel flow up to $Re=11200$.

Built a Bayesian optimization framework to efficiently tune turbulent flow control parameters with 70% compute savings.

PROJECTS

Differentiable biomechanics of swimming

2026

Developed JAX-based differentiable solver and a graph-based descriptor tool for high-dimensional fluid-dynamical landscapes.

EDUCATION

University of Sheffield

UK

PhD, Fluid Mechanics

2016 – 2021

Dissertation: drag reduction through active and passive rotating actuators in wall-bounded turbulent flow

University of Genoa

Italy

B.Sc. and M.Sc., Naval Architecture

2010 – 2016

Exchange semester, Naval Architecture – TU Hamburg, Germany (2015)

PUBLICATIONS & GRANTS

P. Olivucci, DJ Wise, P. Ricco, Reduction of turbulent skin-friction drag by passively rotating discs, J. Fluid Mech. (2021)

P. Olivucci, X. Shao, M. Albers, W. Schröder, R. Semaan, Multi-fidelity modeling of net power savings of an actuated turbulent boundary layer, Comp. & Fluids (2023)

P. Olivucci and D. E. Rival, The search for the gust-wing interaction "textbook", under journal review (2026)

P. Olivucci, D. E. Rival and R. Semaan, Control-oriented cluster-based reduced-order modelling, under journal review (2026)

Co-author of DFG grant proposal currently under review (Germany, 2025);

Co-author of proposals awarded $\sim 2,000,000$ core-hours on the UK National Supercomputer Facility

LANGUAGES

English (fluent), German (C1+), Italian (native), Spanish (conversational)